

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

December 2012

EC 303 Audio Video Engineering

Date: 28.11.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1(a) Define Directivity of Microphone. [02]
- (b) Find the resistance of the amplifier if 25 W output power gives 100 V on the loudspeaker line. What will be the per cent reduction of loss in line if the line resistance is 1 ohm, and the loudspeaker resistance is 16 ohm? [04]
- (c) The sound would be _____ without reverberation in a concert hall. [01]
- Q - 2 (a) Explain the parameters on which quality of a microphone depends. Give the values of these parameters for moving coil, ribbon, condenser, and carbon microphone. [05]
- (b) Sound intensity at 1 metre from a loudspeaker is 400 mW/m^2 . The audio power fed to the loudspeaker is 10 W. calculate the efficiency of the loudspeaker. What type of speaker it could be? [03]
- (c) What are the factors on which reverberation time depends? Give typical values of reverberation time for church or temple, concert hall, auditorium, lecture hall, studio. [06]

OR

- Q - 2 (a) Sensitivity of a microphone is 80 dB below 1 volt for sound pressure of 0.1 Pa. find out its output in microvolts. [05]
- (b) Why cone type speaker is known as direct radiating type speaker? [03]
- (c) What are the requirements for a good auditorium for pleasant listening? Give salient features of acoustical design for an auditorium. [06]
- Q - 3 (a) Find out the intensity of sound in micro watt at a distance of 50 metre from a loudspeaker whose efficiency is 20% and PA system feed 10 watt audio power to the loudspeaker. [05]
- (b) Differentiate Hypermedia and Multimedia with example. [06]
- (c) Explain flat panel display television receivers. [03]

OR

- Q - 3 If normal conversation level is 25 micro watt and we need at least 3dB more for comfortable listening in a meeting, calculate the intensity of sound required in dB over threshold of hearing (1 pW), presuming 3dB of power is absorbed by the audience. [05]
- (a) [06]
- (b) List the application of Multimedia and explain any one. [06]
- (c) Explain three dimensional television pictures. [03]

SECTION - II

- Q - 4 (a) What do you understand by active and blanking periods in horizontal and vertical scanning? Give periods of nominal, active and retrace intervals of horizontal and vertical scanning as obtained in the 625 line system. [04]
- (b) Define Following. [03]
1. D.C. Component of the video signal
 2. Pedestal Height
 3. Blanking Pulses
- Q - 5 (a) How many horizontal lines get traced during each vertical retrace? What is the active number of lines that are actually used for picture information pick-up and reception? [04]
- (b) Sketch composite video signal waveform for at least three successive lines and indicate: (i) blanking level (ii) pedestal height (iii) sync pulse level. [04]
- (c) What is vestigial side band transmission and why it's used for transmission of TV picture signal? [06]

OR

- Q - 5 (a) Justify the choice of 625 lines for TV transmission. Why the total number of lines is kept odd in all television systems? [04]
- (b) Sketch details of horizontal blanking and sync pulses. Label on it (i) front porch, (ii) horizontal sync pulse, (iii) back porch and (iv) active line periods. [04]
- (c) Describe briefly co channel and adjacent channel interference effect. Discuss the techniques employed to eliminate such interference in fringe area. [06]
- Q - 6 (a) What do you understand by image rejection ratio? Explain how by providing an RF amplifier, image signal reception is greatly minimized. What are the other merits of using an RF amplifier before the frequency converter? [06]
- (b) Why local oscillator frequency kept higher than received signal? [04]
- (c) Discuss the merits and de merits of positive and negative amplitude modulation. [04]

OR

- Q - 6 (a) Draw block diagram of a monochrome TV receiver and briefly review the nature of input and output signals at various sections. [06]
- (b) Explain the function of SAW filter. [04]
- (c) Sketch channel bandwidth details and show location of Y, C and sound carrier frequencies. Why is a guard band allowed at the higher end of channel width spectrum? [04]
